FYP-2 Mid Evaluation — 5-Minute Speaking Script

S26-017-R-GASR · FAST NUCES Islamabad · Supervisor: Dr Akhtar Jamil

Written against the finalised 24-slide deck. **All 24 slides are covered.** Slides 1–7 are the spoken talk and the only ones that count against the five minutes; 8–12 carry the live demonstration; 13–24 are answer material for questions.

Member	Slides	Words	Time	Section
Ahmad Rashad 22i-1175	1 – 3	228	1:38 – 1:49	Goals and objectives · faculty feedback addressed
Shehroz 22i-0771	4 – 6	309	2:12 – 2:28	Progress: methodology, what we measured, the base papers
Humair Khan 22i-2623	7	86	0:37 – 0:41	Remaining work, then drives the live demonstration
Total		**623**	**4:27 – 4:59**	Counted from the spoken lines only. Fast end 140 wpm, slow end 125 wpm.

Two rules this script follows. (1) Numbers stay on the slides. A figure is spoken aloud only when it means something on its own — "all ten videos", "one camera instead of two". Error scores, correlations and decibels are on screen where the panel can read them at their own pace, and are described in words instead. **(2) Nothing is read off a slide.** The deck carries the detail; you carry the argument.

Hard constraints from the committee. The 4–5 minutes are **uninterrupted** — do not stop for questions during slides 1–7, and do not overrun; every minute borrowed is a minute taken from the demonstration. Everything on **one laptop**. The demonstration must be the **live system**, not a video.

Ahmad Rashad (22i-1175) — Slides 1 to 3

Slide 1 — From Reconstruction to Anticipation · 28 words · 12 s

Good morning. I'm Ahmad Rashad, with Shehroz and Humair. Our supervisor is Dr Akhtar Jamil. Project S26-017 — a research FYP on 4D surgical reconstruction and deformation prediction.

Slide 2 — What the project is for · 104 words · 45 s

A surgeon operating through one laparoscope sees a flat image of the present moment. Our goal is a system that reconstructs that scene in 3D and anticipates how the tissue will deform next — from one camera, with no depth sensor and no annotation.

Point along the four cards. Say every status — the panel is marking progress.

Four objectives. Reconstruct in 3D from a single camera: complete, that was FYP-1. Learn how tissue deforms, without labels: complete. Predict forward with its own confidence: implemented. Evaluate against the literature: in progress.

Pause. Read the question slowly — it is the sentence they should remember.

So the research question is this. Can future tissue geometry be predicted from ordinary surgical video — and can the system tell us when to trust it?

Slide 3 — Faculty feedback from FYP-1 · 96 words · 41 s

Say the feedback, say what was built. The file names are on the slide — do not read them out.

We had four pieces of feedback after FYP-1, and all four are implemented.

Implement a base paper for comparative analysis — we implemented two; Shehroz will come back to them.

Implement state-of-the-art methods — we built eleven, classical and learned, all measured the same way on the same data.

Do a detailed tuning and optimisation analysis — that is Module 8.

And begin a paper in conference format — there is an IEEE manuscript in the repository.

All four are written. Two still need their compute runs — the first item on our remaining-work slide.

Step back from the laptop.

Shehroz.

Shehroz (22i-0771) — Slides 4 to 6

Slide 4 — The methodology, finalised · 90 words · 39 s

Thank you, Ahmad. The methodology is fixed. We track points on the tissue surface, look at the last few frames of how they moved, and predict where they go next. Nothing in that chain needs a human to label anything.

The model is a diffusion model — the family used for image generation, but generating motion rather than pictures.

The middle card. This is the one decision worth naming.

One decision matters most. Rather than predicting the motion outright, it predicts the *correction* to a simple straight-line guess — so even when it learns little, it still beats guessing naively.

Slide 5 — Metrics chosen, and what they measured · 102 words · 44 s

Lead with the three questions, not the metric names. The figures are on the slide.

We measure three things: is the shape in the right place, is the motion pointing the right way, and does the model know when it is unsure.

The second matters because a shape score alone cannot separate a model that learned nothing from one that learned the right direction but the wrong distance.

Three answer cards, right side. Say what each means, not its value.

The answers. On accuracy we are not yet ahead of the simple baselines, and we lose on all ten videos. But the direction is clearly learned, and the model's confidence tracks its real error. So the failure is *how far*, not *which way* — a far more tractable problem.

Slide 6 — What we build on, and where we go past it · 117 words · 50 s

The comparison slide, and it deserves its time. Four rows, left then right.

Four bodies of work sit next to ours.

EndoGaussian and Deform3DGS reconstruct surgical scenes beautifully — but with two cameras, and only what already happened. We reach comparable quality from one camera, then predict forward.

PointRNN is the closest published method to ours, and our main learned baseline — but built for rigid outdoor scenes, and it gives one answer with no confidence.

Luo and Hu built the first diffusion model over 3D points — but it generates shapes, not futures. Ours makes that idea conditional on time.

And surgical anticipation does predict the future — but in flat pixels, and only when told the instrument's action.

Read the gold line as your closing sentence.

No published method does all of it at once.

Step back.

Humair.

Humair Khan (22i-2623) — Slide 7, then the demonstration

Slide 7 — Remaining work until the final evaluation · 86 words · 37 s

Give each item a status, not a hope.

Five things remain, in the order they have to happen.

One: run the full comparison and the tuning studies. Both are written and tested — what is left is compute, not code.

Two: evaluate the two corrections already implemented. One of them needs no retraining at all.

Three: hold out an entire video, rather than frames from videos we also train on.

Four: fold the finished results into the manuscript.

Five: submit, and grow the dataset.

Go straight into the demo. Do not pause for questions.

And with that, let me show you the system running.

The Demonstration — Humair drives, slides 8 to 12

No verbatim script, because a live system will not run to one. These are the lines to have ready while cells execute.

Slide 8 — while the notebook loads

Everything from here is live — nothing is pre-rendered. We load the trained model, check the data it is being tested on, predict a moment it has never seen, draw that prediction as an actual video frame, compare it against the real one, and then run past the end of the recording.

Slide 9 — while the render cell runs

There is a gap to bridge here. The prediction moves a few hundred tracked points; a picture needs the whole scene, which is tens of thousands. So the motion of the tracked points is carried outward onto everything around them.

Two honest notes about what you are seeing. Tissue moves only a pixel or two between frames, which is invisible, so the arrows are magnified — and that magnification is printed on the panel and never applied to any measurement. And the forecast runs past the end of the recording: the real-video panel stops and says so, while the predicted panel keeps going. That is the only output that shows prediction rather than describing it.

Slides 10 to 12 — describing the results

Agree beforehand whether you or Shehroz narrates these. One of you, not both.

The key finding: the model knows which way the tissue moves, not how far. Then uncertainty — when the model's own samples disagree with each other, its prediction really is worse, which means it can flag its own weak moments. A system that is confidently wrong is unusable in theatre; one that says "don't trust me here" is usable even when imperfect. And finally the full comparison against both simple baselines.

> **Demo checklist — the day before, not on the day.**
>
> - Runtime connected, GPU confirmed, checkpoint verified loadable.
> - Run the whole demo path end to end twice. Time the slowest cell; know what you will say over it.
> - Deck, notebook, checkpoint and outputs on **one laptop**.
> - Fallback: if the runtime dies, show the already-executed notebook and say plainly that it is not re-executing. Do not pretend.
> - Screen resolution set so notebook text is readable from the back of the room.

Questions — and where the numbers live

Because the talk is deliberately light on figures, expect a question asking for them. That is fine: every figure is on a slide. Do not recite from memory — turn to the slide.

If the question is about…	Answers · slide
The network itself, why diffusion, the forward pass, conditioning, training, overfitting	**Ahmad** · 14 (architecture), 15, 16, 19

Goals, scope, how the FYP-1 feedback was addressed	**Ahmad** · 2, 3
Metrics, baselines, the quality tests, evaluation integrity, the comparison numbers, the direction finding, uncertainty	**Shehroz** · 5, 10–12, 17, 18
Literature positioning, the base papers, reconstruction against published figures	**Humair** · 20, 23
The demo, rendering, the forecast, the dataset, the pipeline	**Humair** · 9, 21, 22

Reference Slides 13 – 24 — What to Say If You Jump There

None of this is presented. It is spoken only when a question sends you to the slide, so none of it counts against the five minutes. Each entry is roughly 20–30 seconds — answer, then stop talking.

How to use this section. Do not walk through these slides in order. Jump straight to the one the question needs, give the short answer below, and return. If you do not know which slide holds the answer, the routing table above tells you. Typing a slide number then Enter jumps there in the HTML deck; in PowerPoint, the number then Enter works the same way.

Slide 13 — Reference slides · divider, never spoken

A dark divider. It exists so the panel can see the talk has ended and everything after is answer material. Pass straight over it.

Slide 14 — Architecture, Modules 5 and 6 · Ahmad · the most-used slide here

Triggered by: "how does the model actually work?", "what is the architecture?", "where does the conditioning go?" Trace it left to right with your hand; do not read the boxes.

The top band is one forward pass. Two branches: the observed geometry along the top, the noisy candidate prediction along the bottom. Each is encoded pointwise, then the context goes through a transformer that attends along *time* only — four frames per point, points folded into the batch.

The red box is the one I would point at. The prediction horizon is embedded and added to *both* branches. Without it the same four frames carry three different right answers and the task is unlearnable at any capacity.

The two branches then exchange information in both directions through cross-attention, get concatenated, and a small head decodes the motion. The arrow on the right is the only thing that crosses into Module 6 — a checkpoint file. Module 6 never trains.

Slide 15 — Module 5, learning how tissue moves · Ahmad

Triggered by: "why diffusion?", "what does it take as input?", "why is the horizon an input?"

Input is three things: the last four frames of the tracked points, the horizon, and a patch of noise. Output is the movement of each point — and because we can re-run it with fresh noise, several plausible futures from

the same input.

On why diffusion rather than a single regression: soft tissue under contact is genuinely uncertain, and a single estimate hides that. Sampling repeatedly gives us a spread, and that spread is what becomes the confidence signal on slide 11.

Slide 16 — Module 5, what went wrong and what we did · Ahmad

Triggered by: "what problems did you hit?", "why did you change the target?", "what about overfitting?" Five rows — name the one they asked about, not all five.

Each row is a problem found by measurement and the change that followed. The one worth volunteering is the last: we found that a model which simply gave up scored exactly the same as doing nothing, because predicting zero displacement *is* the naive baseline. So we changed what it predicts — the correction to a straight-line guess rather than the motion itself. Giving up now falls back to something meaningfully stronger than naive.

Slide 17 — Module 6, using the model and measuring it honestly · Shehroz

Triggered by: "what does Module 6 do?", "how do you evaluate?", "what are the quality tests?"

Module 6 trains nothing. It loads the checkpoint and asks three different questions: predict once, predict repeatedly by feeding answers back, and predict five times from the same input to see how much they disagree.

Everything is graded against two deliberately simple rules — nothing moves, and everything continues straight. Our position is that if a model this size cannot beat those, we say so rather than changing the metric.

Slide 18 — Module 6, what went wrong and what we did · Shehroz

Triggered by: "how do you know the evaluation is sound?", "did you validate your baselines?" This slide is your strongest answer to a methodology challenge.

These are mostly about measurement integrity rather than modelling. The first row is the one I would point at: rollout needs continuous trajectories but the upstream module stores fixed windows, so we rebuild the trajectory and then *verify* the rebuild against what was actually saved. It runs as a gate — nothing is predicted until it passes.

The last two rows are about our own baselines. Two of them were wrong when we checked: the constant-acceleration one was taking velocity at the wrong time instant, and what the code called a Kalman filter was really a fixed-gain filter with no covariance. Both were fixed and verified against cases with a known answer. A comparison is only as honest as its baselines.

Slide 19 — Training converged cleanly, and then overfitted · Shehroz

Triggered by: "how did training go?", "did it converge?", "isn't ten videos too few?"

Training itself was clean — smooth convergence, no instability, and it resumes exactly where it stopped across Colab disconnections. What went wrong is that held-out error bottomed out early and then rose while training loss kept falling. That is overfitting, and we measured it directly rather than assuming it. It is

consistent with a corpus of ten short recordings, which is why growing the dataset is on the remaining-work slide.

Slide 20 — Four adjacent fields · Humair

Triggered by: "has nobody done this?", "what is the gap exactly?" Work the columns, not the rows.

Four lines of work sit near ours. Surgical reconstruction gives the 3D geometry but retrospectively. Surgical anticipation predicts the future but in flat pixels, and only when told the instrument's action. Point-cloud forecasting predicts 3D but for rigid outdoor scenes. And medical point diffusion works on tissue but estimates the present state, not a future one.

Our claim is the bottom row, and it is a claim about the combination rather than any single column. Note the last column in particular — none of the four reports uncertainty at all.

Slide 21 — FYP-1 built the reconstruction · Humair

Triggered by: "what did FYP-1 deliver?", "where does this sit in the pipeline?", "what is the reconstruction quality?"

Modules 1 to 4 are complete and validated from FYP-1 — frames and masks, 4D Gaussian reconstruction, trajectory extraction, and the self-supervised training windows. The two in red are FYP-2. The reconstruction runs from a single camera with no depth sensor, and everything downstream was built from it without a human labelling anything.

Slide 22 — The laparoscope shows the present · Humair

Triggered by: "why does this matter clinically?", "what is the motivation?" Only needed if someone challenges the premise.

A surgeon operates through one camera that gives a flat image — no depth, no 3D structure, and no indication of how tissue will respond to the instrument about to touch it. Reconstruction research has solved the first half well and has not addressed the second at all. Every published surgical 4D system is retrospective: it models deformation that has already been recorded.

Slide 23 — How we compare with published work · Humair

Triggered by: "what are your numbers against theirs?", "is your PSNR competitive?" This is where the numeric comparison lives, so bring it up rather than reciting from memory.

On the reconstruction side we sit within a few tenths of a decibel of the stereo state of the art while using a single camera. I want to be careful with that comparison though — those published figures use stereo supervision on different sequences, so it is indicative rather than controlled, and we say exactly that in the paper.

On the prediction side, the two base papers we implemented are listed there. The full eleven-method comparison on one protocol is implemented; the run is the next step.

Slide 24 — Where Modules 5 and 6 stand · anyone · good slide to end on

Triggered by: "so where are you overall?", or use it yourself if the demo finishes with time left. It is also the right slide to be on when questions begin.

Delivered: a complete prediction pipeline needing no clinical labels, a six-test evaluation framework with an

integrity gate, calibrated uncertainty, a comparative study implemented, and a conference paper drafted.

Not yet achieved: beating the naive baseline on accuracy. We are not going to present that as anything else. But we did not leave it unexplained — we localised the cause to displacement magnitude rather than direction, and implemented two fixes that follow directly from that measurement.

If you end here, the last line on the slide is the sentence to say aloud.

An honest negative result with an identified cause is worth more than an unexplained positive one.

The one answer to agree on. "Your model doesn't beat a simple baseline — why should this pass?" One person answers: *"Because we can tell you exactly why it doesn't. We measured that it learns which way tissue moves and fails only on how far; we tested and ruled out the two obvious alternative explanations; and we have implemented the two corrections that follow. We also get a confidence estimate that none of the baselines can give you. For a research project, a measured negative with an identified cause is worth more than an unexplained positive."*